Jinuk Moon

+82-10-7342-3750 | jumoon@snu.ac.kr

jinukmoon | github.com/JinukMoon | jinukmoon.github.io

Seoul National University

Profile

PhD candidate developing machine learning interatomic potentials for accelerating catalyst discovery. Developed CatBench, a systematic benchmarking framework that enables reliable evaluation of MLIPs in heterogeneous catalysis. Research focuses on combining machine learning with first-principles calculations to discover efficient catalytic materials for clean energy applications.

Education

- **Seoul National University** — *2023 – Present*
  Ph.D. Materials Science and Engineering — Seoul, South Korea
  - Machine learning-accelerated computational catalysis for electrocatalyst discovery and design
- **Pohang University of Science and Technology (POSTECH)** — *2019 – 2022*
  B.S. Chemical Engineering — Pohang, South Korea
  - Thesis: Study of Metal-Nitrogen-doped carbon catalysts for Chlorine Evolution Reaction combining DFT calculation with Machine Learning
  - GPA: 3.98/4.3 (Summa Cum Laude)
- **KU Leuven** — *Spring 2022*
  Exchange Student - Faculty of Science — Leuven, Belgium

Open Source & Code

- **CatBench - Benchmark Framework for Machine Learning Interatomic Potentials in Catalysis**
  Systematic evaluation of 15+ MLIP architectures on 47,000+ adsorption configurations
  - pip-installable Python package | catbench.org | github.com/JinukMoon/CatBench

Honors & Awards

- **Presidential Science Scholarship (PhD Track)** — *2025 – Present*
  Ministry of Science and ICT — South Korea
- **Best Oral Award** — *Spring 2025*
  The Korean Institute of Metals and Materials (KIM) — South Korea
  - CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis
- **Best Oral Award** — *Fall 2024*
  The Korean Institute of Metals and Materials (KIM) — South Korea
  - Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential
- **Best Oral Award** — *Spring 2024*
  The Korean Ceramic Society (KCerS) — South Korea
  - Selective control of oxygen evolution mechanisms by doping concentrations in RuO_2
- **Best Poster Award** — *2025*
  The 8th International Conference on Advanced Electromaterials (ICAE 2025) — South Korea
  - Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution

Skills

- **Programming & Frameworks:** Python (Advanced), PyTorch, C/C++ (Intermediate), Bash scripting (Intermediate), HTML
- **Machine Learning:** MLIP Training & Benchmarking (15+ architectures), Dataset Generation, Bayesian Optimization
  - *MLIP Fine-tuning:* Generated 100,000+ in-house DFT calculations, fine-tuned GemNet-OC reducing adsorption energy MAE from 0.48 eV to 0.05 eV (evaluated by CatBench)
  - *Large-scale Molecular Dynamics:* MLIP-accelerated MD simulations (100+ ns) for metal dispersion and structural evolution analysis
  - *High-throughput Screening:* MLIP-accelerated screening of alloy catalysts using bulk structures from Materials Project, OQMD, GNoMe, and OMat datasets
  - *Genetic Algorithm:* Ensemble-GA development with sub-population partitioning and mixing strategies for enhanced global optimization
- **Computational Chemistry:** Density Functional Theory (VASP), Machine Learning Interatomic Potentials (15+ models: UMA, EquiformerV2, GemNet-OC, MACE, etc.), Molecular Dynamics, Electronic Structure Analysis
- **High-Performance Computing:** Supercomputer Cluster Server Management, SLURM, PBS, Batch Job Optimization
- **Research Expertise:** Heterogeneous Catalysis, Single Atom Catalysis, Electrocatalysis (OER, HER, CER), Multiscale Simulation using MLIP, MLIP Benchmarking, ML Analysis of Experimental Characterization
- **Language:** Korean (Native), English (Fluent)

Publications (9 published, 2 under review, 6 first-authored)

† Equal contribution, * Corresponding author

[under review] Kim, J.†, **Moon, J.**†, Cho, K.†, Kim, J.†, Han, M. H., Kim, S., Han, J., Choi, D., Jun, H., Jang, J. H., Oh, H.-S., Cho, S. K.*, Kim, W.*, Han, J. W.*, Lee, J.* (2026). Size-Dependent Dynamic Surface States of Ru Clusters: Modulating Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution Reaction. *Energy & Environmental Science*, under review.

[under review] Jun, H.†, Kwon, D.†, **Moon, J.**†, Han, S.†, Kim, S., Yi, S. Y., Kim, Y., Kang, E., Choi, C. H.*, Han, J. W.*, Kim, W.*, Lee, J.* (2026). Synergistic Effect of Tuning Reaction Mechanism and Interfacial Water for Efficient Acidic Water Oxidation. *Advanced Materials*, under review.

[1] Lim, H.†, Choung, S.†, **Moon, J.**, Han, J. W.* (2026). Angular relational knowledge distillation of machine learning interatomic potentials for scalable catalyst exploration. *npj Computational Materials*, in press.

[2] Park, K.†, Lee, W.†, Park, J., Jo, M., Bae, Y., Kim, M., Shin, S., **Moon, J.**, O'Hayre, R.*, Han, J. W.*, Song, S.*, Park, J. Y.* (2026). Oxygen electrode reaction kinetics of $BaCe_{0.7}Zr_{0.1}Y_{0.1}Yb_{0.1}O_{3-\delta}$ based composite air electrodes for reversible solid oxide cells. *Nature Energy*, in press.

[3] Choung, S.†, Kim, M.†, **Moon, J.**†, Han, J. W.* (2025). From Atomic Motif to Realistic Single Atom Catalysts through Machine Learning Interatomic Potentials. *ACS Energy Letters*, 10, 6288-6296.

[4] **Moon, J.**, Jeon, U., Choung, S., Han, J. W.* (2025). CatBench Framework for Benchmarking Machine Learning Interatomic Potentials in Adsorption Energy Predictions for Heterogeneous Catalysis. *Cell Reports Physical Science*, 6, 102968.

[5] Jung, H.†, Lee, Y.†, Cho, K.†, Kim, T. Y., Song, J., Jung, H. S., Park, S. I., Lee, Y., **Moon, J.**, Park, W., Nam, J., Park, S., Kim, W., Han, J. W.* (2025). Strategic Design of Oxophilic Dopants for Active and Durable Alkaline Hydrogen Evolution Reaction Under Seawater. *Advanced Functional Materials*, e16387.

[6] Jun, H.†, Kang, E.†, **Moon, J.**†, Kim, H., Han, S., Choung, S., Kim, S., Yi, S. Y., Kang, E., Choi, C. H.*, Han, J. W.*, Lee, J.* (2025). Quantity effect of heteroatom incorporation on the oxygen evolution mechanism in ruthenium oxide. *Chem*, 11(5), 102367.

[7] Choung, S.†, Park, W.†, **Moon, J.**†, Han, J. W.* (2024). Rise of machine learning potentials in heterogeneous catalysis: Developments, applications, and prospects. *Chemical Engineering Journal*, 494, 152757.

[8] Choung, S.†, Yang, H.†, **Moon, J.**, Park, W., June, H., Lim, C., Han, J. W.* (2024). Theoretical tuning of local coordination environment of metal-nitrogen doped carbon catalysts for selective chlorine-evolution reaction. *Catalysis Today*, 425, 114358.

[9] Choung, S., Kim, Y., **Moon, J.**, Roh, J., Hwang, J., Han, J. W.* (2023). Unveiling the catalyst deactivation mechanism in the non-oxidative dehydrogenation of light alkanes on Rh(111): Density functional theory and kinetic Monte Carlo study. *Catalysis Today*, 411-412, 113819.

Conference Presentations

[1] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.

[2] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Graph Neural Network (Oral Presentation). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.

[3] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *The 8th International Conference on Advanced Electromaterials (ICAE 2025)*, Jeju, South Korea.

[4] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *Korean Institute of Metals and Materials (KIM)*, Gwangju, South Korea.

[5] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Fine-tuned Machine Learning Interatomic Potential (Oral Presentation). *Korean Institute of Chemical Engineers (KICHE)*, Jeju, South Korea.

[6] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.

[7] **Moon, J.** (2025). CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.

[8] **Moon, J.** (2024). Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, PhyeongChang, South Korea.

[9] **Moon, J.** (2024). Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential (Oral Presentation). *23rd Korea Japan Students' Symposium*, Seoul, South Korea.

[10] **Moon, J.** (2024). Selective Control of Oxygen Evolution Reaction Mechanisms by Adjusting Mn Doping Concentrations in RuO_2 (Oral Presentation). *The Korean Ceramic Society (KCerS)*, Busan, South Korea.